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Monetary policy stance and foreign currency lending: evidence from a persistently dollarized emerging market

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Abstract
This study explores how central bank policy changes influence Turkish banks' foreign currency lending in one of the most dollarized emerging markets. Using quarterly data from 2002Q4 to 2024Q3 and applying a Bayesian Additive Regression Trees model within a machine learning framework, we uncover that higher policy rates amplify foreign currency lending—especially among larger banks—thereby weakening monetary transmission. The analysis also highlights the nonlinear and asymmetric effects of deposit dollarization, bank size, and macroeconomic conditions. These findings offer new insights into the structural constraints of monetary policy under persistent dollarization, with direct implications for policy design in similar economies.

Keywords Machine learning · Dollarization · Türkiye · Monetary policy · Policy rate · Foreign currency lending

JEL Classification E5 · F31 · G21

1 Introduction

Dollarization refers to the widespread use of foreign currency alongside the domestic currency in activities such as daily transactions, savings, and lending. Dollarization has been a prevalent phenomenon in several emerging countries including Türkiye. It is one of the peculiar characteristics of Turkish economy that has been on the scene for the past five decades. Despite fluctuations in its level throughout the course of this time period, it has remained very resilient.

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Although extensive literature exists on dollarization, the majority of these studies focus on the drivers of dollarization (Ize and Levy Yeyati 2003). Dollarization has enormous potential to paralyze conducting effective economic policies. In the realm of monetary policy, central banks' operational capacity to pursue financial market stability, promote growth and control inflation may be impaired by dollarization. Despite the substantial potential of dollarization to distort the efficacy of monetary policy measures, it has not received considerable research attention.

Bank lending behavior is particularly susceptible to the complex risks introduced by dollarization. Under standard monetary transmission theory, contractionary policy—such as an interest rate hike—typically reduces bank lending, with stronger effects observed among small and illiquid banks. However, in a dollarized financial system, this relationship becomes significantly more complex. Dollarization adds new dimensions of risk to both the asset and liability sides of banks' balance sheets, resulting in heterogeneous responses to monetary tightening. In particular, when banks hold substantial volumes of foreign currency deposits and loans, the intended effects of contractionary policy may be weakened or distorted. Understanding how the credit channel operates under such conditions is therefore critical. Yet, to date, there appears to be a lack of empirical studies that examine how dollarization shapes foreign currency lending behavior in response to monetary policy—especially in the context of Türkiye.

A significant gap in the existing literature motivates this investigation. Due to this gap, it is crucial to carry out analyses aiming at this issue. Therefore, the objective of this inquiry is to carry out an in-depth examination of the importance of bank lending behavior in a dollarized environment in Türkiye. It focuses on the bank lending behavior and examines the impact of monetary policy adjustments by the Central Bank of the Republic of Türkiye (CBRT, henceforth) on the behavior of commercial banks in a significantly dollarized environment. In order to accomplish this objective, the study executes two distinct analyses. The first part of the analysis examines whether banks respond to monetary tightening or easing by shifting their lending portfolios toward foreign currency loans at the expense of domestic currency credit. In the second stage, we assess the robustness of these findings by analyzing the broader determinants of bank loan growth. This complementary analysis evaluates the role of foreign currency lending in explaining variations in total bank credit growth in Türkiye. Additionally, the study explicitly distinguishes between small and large commercial banks to account for potential heterogeneity in behavior across institutions of different scales.

In order to fulfill these objectives, the methodological tool employed by this study is Bayesian Additive Regression Trees (BART, *henceforth*) algorithm operating under machine learning (ML, *henceforth*) framework. The dataset utilized in this study covers the period from 2002Q4 to 2024Q3. In the above-mentioned analyses, the share of foreign currency loan and the growth in banks' total loan are used as dependent variables in the first and the second analysis, respectively. The study uses policy interest rate as a variable reflecting CBRT's policy stance. On the explanatory side, the study incorporates both bank-specific factors (deposit dollarization, size, liquidity, profitability) and also control variables (inflation, GDP growth, real exchange rate, trade openness).

BART was chosen as the major machine learning methodology in this work for both methodological and empirical reasons. BART is distinguished from tree-based approaches by its nonparametric Bayesian foundation, which enables the flexible modeling of complicated, nonlinear interactions without necessitating that the researcher pre-specifies functional forms or interaction structures. BART is particularly well-suited for financial applications in which macroeconomic indicators, bank-level characteristics, and policy instruments interact in complex and dynamic ways, as it efficiently captures higher-order interactions and subtle nonlinearities, in contrast to conventional regression models. In dollarized economies, the reaction of foreign currency loans to monetary policy shocks is extremely nonlinear and asymmetric.

The ML approach employed in this study is implemented in two phases. In the first phase, "the variable importance ranking" (VIR, *henceforth*) method is utilized to quantify and rank the explanatory variables based on their relative influence on banks' lending decisions. In the second phase, "partial dependence plots" (PDPs) are used to visualize and interpret how key factors affect banks' foreign currency lending behavior and the overall growth of total loans. This two-step framework enables a more nuanced understanding of the underlying relationships beyond what traditional linear models typically reveal.

This study makes a number of novel contributions to the existing literature. First, to the best of our knowledge, no previous research has investigated how central bank policy changes affect banks' foreign currency lending behavior in the context of Türkiye. The only closely related work is Mora (2013), which explores a similar question for Mexico using conventional panel econometric techniques. In Türkiye's case, a few relevant studies exist (e.g., Rengifo et al. 2013; Özsöz and Algan 2015; Çağlayan and Talavera 2016; Yılmaz 2020; Çağlayan et al. 2021), but they differ significantly in both scope and methodology. Most notably, this study is the first to apply a ML framework—specifically the BART algorithm—to this area of inquiry. Second, while earlier studies mainly focus on identifying the determinants of foreign currency lending—often emphasizing households' tendency to hold dollar-denominated accounts under risk—our analysis takes a broader perspective by examining how dollarization influences the overall structure of bank lending portfolios. By addressing both the causes and the consequences of foreign currency lending, the study provides a more nuanced understanding of how dollarization reshapes the transmission of monetary policy within the banking sector. Third, the dataset used in this study provides a significant advantage in terms of time coverage. While earlier research mainly focused on the 2002–2015 period—when dollarization in Türkiye was generally declining—our dataset extends up to 2024, capturing the renewed rise in dollarization that began after 2016. This resurgence was driven by a series of macrofinancial shocks, most notably the 2018 Brunson crisis, which caused a sharp depreciation of the Turkish lira and a shift in currency substitution behavior (Ulug et al. 2023). By covering both the decline and resurgence phases, the study traces a U-shaped path in the evolution of dollarization over two decades. This extended coverage allows for a more dynamic analysis of how changing macrofinancial conditions influence bank lending behavior in a persistently dollarized environment. Finally, the study makes a methodological contribution by applying a

ML framework—specifically the Bayesian Additive Regression Trees (BART) algorithm—to analyze credit dollarization dynamics. To our knowledge, this represents the first application of BART in the context of monetary transmission in a financially dollarized economy. The model's nonparametric, interaction-rich structure allows us to uncover nonlinear and threshold-driven effects that are especially relevant in settings where foreign currency lending behavior reflects strategic responses to macroeconomic and policy uncertainty. These features are central to understanding the transmission of monetary policy under persistent dollarization. Recent research (e.g., Mullainathan & Spiess 2017; Sirignano et al. 2018; Albanesi & Vamossy 2019) supports the idea that ML tools are particularly effective in detecting structural asymmetries and behavioral heterogeneity that standard econometric models often miss.

The study is organized into seven sections. Section 2 provides an in-depth overview of the Turkish experience with dollarization and the structure of the banking sector. Section 3 reviews the theoretical and empirical literature relevant to credit dollarization. Section 4 presents the data sources and methodology, including the use of the BART algorithm. Section 5 reports and interprets the empirical results. Section 6 offers a comprehensive discussion of the findings, linking the empirical patterns to underlying economic mechanisms, macroeconomic conditions, and policy implications. Finally, Sect. 7 concludes the study and outlines key policy recommendations.

2 Background about dollarization and banking sector in Türkiye

2.1 About dollarization experience and monetary policy

Before 1980, despite significant economic challenges and instabilities, such as high and volatile inflation, Türkiye did not face the issue of dollarization. The main reason for this was the prohibition of foreign currency accounts, which restricted the public's access to foreign currencies (Erdem and Yücel 2015). Türkiye implemented significant deregulation steps and liberalization reforms beginning in 1980. The restrictions on foreign currency accounts were completely removed in 1989. The removal of restrictions led to a massive boost in the demand for foreign currency deposits. Note, however, that the increase in dollarization during the late 1980s and throughout the 1990s stemmed significantly from economic crises, ineffective stabilization measures, and limited advancement in capital markets (Metin-Özcan and Us 2007). The percentage of foreign currency deposits in total deposits was less than 10% during the mid-1980s, but it increased to 58% during the financial crises of 1994 and 2001 (Akinci 2009).

As clearly expressed in its charter, CBRT is responsible for the maintenance of economic and price stability. Nevertheless, the CBRT's capacity to pursue these objectives was impeded by the frequent interference of incumbent political parties prior to 2002. The 2001 financial crisis had a devastating impact on the Turkish economy and society, which led to the implementation number of structural and institutional reforms. One of these reforms was the adoption of inflation targeting.

However, incumbent politicians have frequently regarded this rule-based system unfavorably as a result of its restrictive nature. The inflation targeting strategy was initially implemented in 2002 and was fully operational by 2006. Nevertheless, it is important to note that the CBRT has encountered direct or indirect political interference on numerous occasions since 2006. In other words, the CBRT continued to employ monetary interventions to address immediate economic concerns, despite the full implementation of inflation targeting in 2006, illustrating the ongoing struggle between the demands of short-term economic management and rule-based policy adherence (Başçı et al. 2007; Akinci 2009).

After 2001, the dollarization outlook has softened. In particular, due to those structural reforms implemented after the 2001 financial crisis and also prudent economic management, a remarkable performance observed in price stability where inflation decreased from 54.4% in 2001 to 6.3% in 2009 (CBRT 2020). During the same period, however, dollarization was not entirely eliminated, supporting a strong hysteresis component in dollarization. In particular, in 2002, the dollarization rate was approximately 58%, and it reached its lowest point of approximately 30% in 2011.[1]

In the aftermath of 2015, both the real and financial sectors experienced significant distress due to increasing uncertainty. It is important to emphasize that the winding down of expansionary quantitative easing (QE) policies enacted by US and European Union financial authorities following the 2008 Global Financial Crisis in late 2014 had a significant impact. The global contractionary cycle after 2015 led to fluctuations in global financial markets and capital outflows from emerging economies, which in turn affected the financial stability of these countries, including Türkiye.

After 2015, the growing unease was driven by risks emanating from significant political tensions within the domestic sphere and geopolitical developments, including the COVID-19 pandemic, conflicts and wars. Among the major turning points that accelerated the dollarization trend in Türkiye was the so-called Brunson crisis of 2018, which triggered a sharp depreciation of the Turkish lira and contributed to persistent inflation. These developments laid the groundwork for a new policy direction, culminating in the Türkiye Economy Model. As highlighted by Ulug et al. (2023), these episodes reshaped the context in which monetary policy is conducted and perceived. The combination of those factors generates instability, which subsequently undermined inflation by influencing economic confidence and expectations. The COVID-19 pandemic not only created the fiscal vulnerabilities (OECD 2022) which increased the pressure on government finances but also created supply chain disruptions around the world. Aftermath of the pandemic, Russia's military conflict with Ukraine posed additional strains to the Turkish economy (IMF 2022). All these developments brought about significant increases in global commodity prices, particularly oil and natural gas (OECD 2022). Due to these undesirable developments,

[1] As a side note, while households have provided virtually all foreign currency deposits in the banking system during the past forty years, foreign currency lending to households was banned after 2009 and this regulation is still intact.

the Turkish Lira has lost substantial value against other currencies and this trend gained momentum after August of 2018. The Turkish Lira's significant depreciation against other currencies has fueled inflation (due to standard pass-through mechanism) and intensified economic instability. This has led to a feedback loop, exacerbating the overall economic instability. Economic agents are increasingly resorting to foreign currencies to hedge against volatility and loss of confidence in the domestic currency, further entrenching the phenomenon of dollarization within the Turkish economy. In 2022, consumer inflation soared to an exceptionally high level of 80%. In 2013, 33% of the deposits were in foreign currency. By 2018, this percentage had increased to 47%, and by 2023, it had reached around 58%. These adverse dynamics were fueled not only by worsening geopolitical risks but also, as Ulug et al. (2023) emphasize, by the adoption of ultra-loose monetary policies, which collectively undermined macroeconomic stability and reinforced the resurgence of dollarization in Türkiye.

2.2 Turkish banking sector

In numerous emerging economies, such as Türkiye, financial markets are frequently marked by underdeveloped capital markets. This has resulted in a financial architecture that is heavily reliant on banks, particularly on deposit banks (Tuna and Almahadin 2021; Mashamba 2022). This characteristic is exemplified by the Turkish system, as the Bank Association of Türkiye reported in 2023 that banks hold an 83 percent share of the finance industry by asset size (BAT 2023). Not only Turkish financial system is dominated by commercial banks, but the aggregate loan by these commercial banks is mainly financed by deposits (Alper et. al., 2014).

In addition to the pivotal role played by banks, several other features of the Turkish banking industry deserve further clarification. Firstly, the presence of a large number of banks with diverse ownership patterns is a notable characteristic. The sector in 2024 consists of a total of 52 banks (TBB 2024). The banks operating in the sector may be categorized into three distinct groups: deposit banks, investment and development banks, and Islamic banks. The number banks operating for each category is 34, 13 and 5, respectively. Secondly, the deposit banks are at the core in this bank-centered system. To clarify further, the share (%) of assets in total banking sector for each group is: deposit banks (86%), investment and development banks (8%), and Islamic banks (6%) (BAT 2023).

Thirdly, among these 52 banks, the ownership structure is also important: 21 banks are owned by foreign entities, and 3 banks are owned by the state. Finally, the banking system exhibits a significant level of concentration. While the combined market share of the three leading public banks is approximately 40% for total deposits and 45% for total loans, the three leading private banks hold around 30% of total deposits and 35% of total loans (BRSA 2021). To put it differently, the top six deposit banks (both public and private) collectively hold approximately 70% of all deposits, highlighting the significant concentration in which a small number of large banks have control over an important portion of the market.

3 Literature review

Understanding how monetary policy affects bank lending in a dollarized environment has been a central theme in macrofinancial research. In these settings, dollarization does not merely distort price signals; it introduces what Calvo and Reinhart (2002) describe as an additional layer of complexity into the monetary transmission process. Reinhart et al. (2003) similarly argue that liability dollarization can transform exchange rate risk into default risk, thereby impairing financial stability. Disyatat (2011) finds that across countries, the transmission from monetary stance to loan supply is highly heterogeneous due to structural and regulatory factors. These frictions are further intensified in dollarized economies.

A key mechanism frequently cited in the literature is currency-sensitive lending, whereby banks adjust their loan portfolios toward foreign currency lending in response to shifts in their deposit structures. Mora (2013), in a benchmark study on Mexico, shows that banks with higher levels of FX liabilities respond to contractionary monetary policy by increasing FX lending, as a form of portfolio rebalancing. This behavior is not idiosyncratic. Brown and De Haas (2012) show that banks across Central and Eastern Europe behave similarly, with deposit dollarization acting as a major determinant of foreign currency loan issuance. Crespo Cuaresma et al. (2011) further argue that bank size, capitalization, and access to external funds intensify these responses, particularly in environments of monetary tightening.

In addition to bank-level variables, macroeconomic and structural factors also shape these outcomes. Rosenberg and Tirpák (2008) and Luca and Petrova (2008) document the role of natural hedging behavior in export-oriented firms, who borrow in foreign currencies to match their revenue streams. In contrast, Edwards and Magendzo (2003) show that high levels of dollarization increase the sensitivity of domestic credit markets to US interest rate cycles. Arteta (2005), Ize and Levy-Yeyati (2003), and Zettelmeyer et al. (2010) emphasize that inflation and exchange rate volatility amplify dollarization incentives, making the credit channel more unstable.

The Turkish economy provides a unique case to study these dynamics. Following the 2001 financial crisis, reforms significantly reduced financial fragility and improved monetary policy effectiveness (Başçı, Kara, & Öğünç, 2007). However, since 2016, Türkiye has experienced macroeconomic instability marked by rising inflation, policy uncertainty, and a decline in central bank independence. The 2018 Brunson crisis, followed by the adoption of the Türkiye Economy Model in 2021—which prioritized low interest rates despite rising inflation—has substantially weakened monetary credibility (Ulug et al. 2023). These events have reignited both deposit and loan dollarization, shifting bank behavior away from traditional responses to policy rates.

Several studies have addressed parts of this transformation. Özsöz and Algan (2015) empirically demonstrate that a 1% increase in deposit dollarization can raise the FX loan ratio by up to 0.66%. Yılmaz (2020) highlights the importance of noncore FX liabilities and firms' hedging motives. Çağlayan and Talavera

(2016) show that banks with tighter liquidity positions are more likely to lend in foreign currencies during policy tightening. Earlier studies by Aktaş and Taş (2007), Sengonul and Thorbecke (2005) and Özsuca and Akbostancı (2012) provide mixed evidence on how size, liquidity and capitalization affect the loan supply reaction to monetary policy, but generally support the presence of a bank lending channel in Türkiye. Özsuca (2022) confirms that these dynamics have strengthened in the post-2011 period.

Despite these valuable contributions—including quantitative linkages between deposit and loan dollarization (Özsöz & Algan 2015), firm-level insights into hedging behavior (Yılmaz 2020) and bank-specific monetary responses (Çağlayan & Talavera 2016)—notable gaps remain in the literature. First, most of these studies focus on the determinants of foreign currency lending rather than on how changes in the monetary stance directly alter the composition of bank credit. Second, nearly all of the existing work is limited to the 2002–2015 period, when dollarization in Türkiye was declining. They overlook the sharp resurgence of dollarization and institutional weakening after 2016. Third, conventional regression techniques used in this literature are generally linear, whereas recent findings—including those in this study—suggest that credit dollarization responds to monetary shocks in highly nonlinear and threshold-dependent ways.

This study contributes to the literature in three primary ways. First, it covers the entire U-shaped evolution of dollarization in Türkiye, from the post-crisis decline (2002–2015) to the recent resurgence (2016–2024). Second, it employs Bayesian Additive Regression Trees (BART), a flexible machine learning algorithm well-suited for detecting nonlinear and asymmetric effects. Third, it explicitly analyzes the heterogeneous behavior of large and small banks in reallocating credit across currencies in response to monetary shocks. By doing so, it provides a comprehensive picture of how dollarization interacts with institutional asymmetries to constrain the transmission of monetary policy in Türkiye and, by extension, in other partially dollarized emerging markets.

4 Data, variables, methodology and empirical models

4.1 Data and variables

The dataset includes 19 commercial banks in Turkish banking industry covering the period over 2002Q4 and 2024Q3. In our empirical research, we use a large dataset including macroeconomic or control indicators and bank-specific factors. This use of an extensive dataset is crucial to our inquiry of the determinants of loan dollarization in commercial banks under different monetary policy stances. The following is an outline of the variables covered in our study (Table 1):

Primary dependent variables in our study are the loan dollarization ratio (loan_dol), which represents the share of foreign currency loans to total loans, and the total loan growth (grwt_tot_loans), which indicates the annual expansion rate of all loans to capture lending trends in the banking sector. This allows us to better understand the factors that influence loan dollarization in commercial banks. The data or

Table 1 Description of the dataset

Variable name	Explanation	Data source
loan_dol	Loan Dollarization Ratio	TBA-Data Query System
grwt_tot_loans	Growth in Total Loans	TBA-Data Query System
dep_dol	Deposit Dollarization Ratio	TBA-Data Query System
Roa	Profit/Total Assets	TBA-Data Query System
Cap	Capital/Total Assets	TBA-Data Query System
Liq	Liquid Assets/Total Assets	TBA-Data Query System
Ms	Market Share (in terms of Assets)	TBA-Data Query System
Rer	Change in Real Effective Exchange Rate	CBRT-EDDS
gdp_grwth	GDP growth rate	CBRT-EDDS
Inf	Inflation rate	IMF-IFS Dataset
Mp	Monetary policy rate	IMF-IFS Dataset
Trade	Trade Openness (Exports + Imports/GDP)	CBRT-EDDS

TBA is Turkish Banking Association, EDDS is the Electronic Data Delivery System provided by the Central Bank of the Republic of Türkiye

variables utilized in the study can be classified into two sets: (1) financial sector and bank-specific variables and (2) control or macroeconomic variables. Financial sector and bank-specific factors are: monetary policy stance (mp), the deposit dollarization ratio (dep_dol), which mirrors whether loan dollarization is affected by the level of the foreign currency funding; the banks' return on assets (roa), which reflects profitability and its influence on foreign currency lending risk; the capital ratio (cap) and liquidity ratios (liq), which are necessary for evaluating financial stability, risk, and short-term solvency; market share (ms), an asset-based indicator of a bank's competitive stance. Exogenous factors or control variables are: The real effective exchange rate (rer), which highlights the impact of currency depreciation on foreign currency lending; GDP growth rate (gdp_grwth), inflation (inf), monetary policy stance (mp), and trade openness (trade). Both set of factors are assumed to collectively influence banking dynamics, interest rates, and price stability.

We include all of these variables in our analysis with their first lagged values so that we can fully understand how loan dollarization works. We adopt first-order lags for all explanatory variables in our empirical models for theoretical and empirical reasons. Since banks adapt their balance sheets slowly to monetary policy and macroeconomic conditions, the bank lending channel usually functions with a lag. Lagged explanatory variables explain for the temporal delay between policy or macroshocks and bank lending choices. Lagged values reduce reverse causality between regressors and dependent variables, minimizing endogeneity problems in econometrics. Using lag structures to simulate dynamic adjustments is widespread in empirical banking and macrofinance literature (e.g., Mora 2013).

This thorough analysis makes use of information from central bank reports, bank-specific financial statements, and international financial databases to provide a detailed investigation of the factors influencing loan dollarization in a variety of monetary and economic contexts.

4.2 Methodology

Machine learning is a branch of AI concerned with developing algorithms that can autonomously learn patterns in data and utilize that knowledge to make predictions or judgments, rather than relying only on human programming. Decision trees are among the most popular and user-friendly machine learning algorithms.

Decision trees are supervised learning algorithms that are nonparametric and are used for both classification and regression applications. A decision tree is composed of nodes that represent characteristics, branches that represent decision rules, and leaf nodes that indicate outcomes. The tree iteratively divides the data into branches depending on feature values in order to optimize the separation between classes or reduce the prediction error in regression.

Bayesian Additive Regression Trees (BART) is a nonparametric Bayesian model that offers a versatile solution to classification and regression issues. Chipman et al. (2010) were the first to present it, and it has subsequently been applied to a variety of domains, including as ecology (Carlson 2020), economics (Clark et al. 2022) and machine learning (Hill et al. 2020).

The BART model can be formulated as follows:

$$y_i = f\left(x_i\right) + \varepsilon_i$$

where y_i is the response variable, x_i is a vector of predictor variables, f is the unknown function modeled as a sum of regression trees, and ε_i are i.i.d. errors typically assumed to be normally distributed: $\varepsilon_i \sim N\left(0, \sigma^2\right)$.

The function f is represented as a sum of m regression trees:

$$f(x) = \sum_{j=1}^{m} g\left(x; T_j M_j\right)$$

where each T_j denotes a tree structure and M_j are the terminal node parameters associated with T_j.

BART incorporates prior distributions to regularize the model: A prior that favors smaller trees (T_j) to prevent overfitting, typically specified as $\alpha(1+d)^{-\beta}$ where d is the depth of the node and α and β are hyperparameters (Chipman et al. 2010). Given the tree structure, terminal node parameters (M_j) are assigned a conjugate normal prior $\mu_{jk} \sim N\left(\mu, \tau^2\right)$. The error variance is assigned an inverse Chi-squared prior $\sigma^2 \sim Inv - \chi^2(v, \lambda)$.

In BART, posterior inference is the process of sampling from the posterior distribution of the trees and their parameters using Markov Chain Monte Carlo (MCMC) techniques. BART suggests modifications to tree structures (e.g., growth, pruning, substitution or modification) and determines whether to approve or reject them. The error variance is updated from its conditional posterior distribution, and the terminal node parameters are sampled from their conditional posterior in accordance with the tree structure.

The regularization and interpretation of the model are significantly influenced by the key hyperparameters in BART, which are k, β and α. The prior probability

of tree depth and structure is determined by the parameters α and β. Shallow trees that limit the danger of overfitting are promoted by $\alpha = 0.95$ and $\beta = 1$ (or 2). This is especially crucial in light of the moderate panel size and rich covariate structure in our data. The prior variance of the terminal node predictions is largely influenced by the k parameter, which also affects the degree of smoothness in the fitted function. A smaller k value results in smoother fits, which is helpful for modeling policy impacts that may grow gradually rather than abruptly. These values were selected in accordance with the current literature (Chipman et al. 2010; Hill et al. 2020) and calibrated through early validation experiments to guarantee interpretability, stability and convergence. Most importantly, the empirical results demonstrate comparable trends across a variety of bank sizes, therefore verifying the robustness of our BART design in the face of variations in model structure and complexity.

BART provides several methodological improvements over conventional tree-based approaches. BART enhances predicted accuracy and minimizes variation by averaging several trees. The Bayesian framework offers a measure of uncertainty by providing reasonable ranges for predictions. BART has the capacity to automatically identify pertinent predictors, which eliminates the need for human feature selection. Complex, nonlinear connections may be modeled using BART without the need for explicit interaction definition.

4.3 Empirical models

As mentioned in earlier parts, our research looks at how monetary policy rates, bank-specific characteristics and macroeconomic variables influence commercial banks' loan dollarization, with a special emphasis on the lending composition of banks within various monetary policy stance. We explore whether monetary tightening prompts banks to prefer foreign over domestic currency loans, potentially diminishing the effectiveness of monetary policy rate on bank loans. Additionally, we examine the influence of factors such as deposit dollarization, bank size, liquidity, capitalization, profitability and macroeconomic indicators on loan dollarization, alongside their interactions amid monetary policy adjustments and the implications for exchange rate risk transference to customers.

To empirically test our hypotheses, we have constructed two economic models. The first model evaluates the impact of the monetary policy stance on loan dollarization, considering both bank-specific attributes (like deposit dollarization, size, liquidity, capitalization and profitability) and macroeconomic variables. It aims to uncover the direct relationship between monetary policy adjustments and the propensity of banks to engage in foreign currency lending. Additionally, this model introduces a novel segmentation by categorizing banks into small and large based on their market shares,[2] enabling us to explore whether the impact of monetary policy on loan dollarization differentiates across different bank sizes.

[2] We classify banks based on their median market share within our data span. Banks in the first quartile (5 out of 19 commercial banks) are labeled as small banks, while those in the last quartile (5 out of 19 commercial banks) are identified as big banks.

The first model can be illustrated as follows:
Model 1:

$$loan_{dol} \sim mp, dep_{dol}, roa, syr, liq, ms, rer, gdp_{grwth}, \text{inf}, trade$$

The second model extends our analysis to the growth of total bank lending in the Turkish commercial banking sector, scrutinizing how monetary policy, alongside loan dollarization and other relevant attributes, influences overall lending activity. This comprehensive approach allows us to assess not only the immediate effects of monetary policy on loan dollarization but also its broader implications for bank lending growth within a dynamic economic landscape.

The second model can be illustrated as follows:
Model 2:

$$grwt_{tot_{loans}} \sim mp, loan_{dol}, dep_{dol}, roa, syr, liq, ms, rer, gdp_{grwth}, \text{inf}, trade$$

Our empirical model is constructed on four core specifications/assumptions:

(1) we include lagged explanatory variables to capture banks' delayed balance-sheet responses and to mitigate endogeneity (Mora 2013; Özsuca & Akbostancı 2012; Brooks 2007);
(2) we allow for nonlinear and potentially asymmetric effects of monetary policy on foreign-currency lending in dollarized markets (Calvo & Reinhart 2002; Arteta 2005; Basso et al. 2011);
(3) we integrate both bank-level controls (e.g., capital adequacy, profitability) and key macroeconomic factors for a comprehensive specification; and
(4) we segment banks into "small" and "large" by market share to explore size-based heterogeneity in monetary transmission.

The use of lagged explanatory variables, for instance, relies on the well-established idea that banks respond to monetary and macroeconomic conditions with a delay due to balance-sheet rigidities and information frictions, which also helps address endogeneity concerns (Mora 2013; Özsuca and Akbostancı, 2012; Brooks 2007). In parallel, the assumption that the response of foreign currency lending to monetary policy is nonlinear and possibly asymmetric reflects dynamics specific to dollarized economies, where risk perceptions, interest rate differentials, and threshold effects shape bank behavior in complex and sometimes counterintuitive ways (Calvo and Reinhart 2002; Arteta 2005; Basso et al. 2011). Given these intricacies, the use of a flexible, nonparametric approach like BART becomes particularly valuable, as it allows for the detection of high-dimensional interactions and non-monotonic patterns without requiring rigid assumptions about the functional form of relationships (Chipman et al. 2010; Hill, 2020), which is crucial in capturing the interplay between variables such as deposit dollarization, market share, capital adequacy and macroeconomic volatility (Levy-Yeyati, 2006; Gomez-Gonzalez and Kutan, 2021). Moreover, the decision to segment banks by size is based on evidence that larger banks tend to operate under structurally different conditions—having broader access to funding, greater exposure to foreign currency and differentiated

policy sensitivity—compared to smaller counterparts (Fungáčová et al. 2014; Özsöz and Algan 2015; Brooks 2007), and incorporating this dimension allows for a more realistic representation of monetary transmission mechanisms. Altogether, these assumptions constitute a theoretically informed and empirically motivated foundation for the model and align with the Ulug et al. (2023) structure of the Turkish banking system as well as broader patterns observed in emerging market economies.

5 Empirical results

In our empirical analysis, we utilize a Bayesian Additive Regression Trees (BART) model to evaluate a comprehensive dataset spanning from 2002Q4 to 2024Q3. This study's empirical specification is based on these two benchmark models, which are designed to thoroughly analyze the underlying relationships in the dataset. Note also that the results for the respective benchmark model differentiate outcomes for small, large and the overall banking sector, as we categorized the banks into two groups based on their sizes: small and large.

The empirical results section includes a comprehensive review conducted through a set of diagnostic tests to confirm the models' robustness and reliability. In addition, the variable importance results are given with great attention to detail, emphasizing the most significant predictors and their relative impact on the model's performance. In order to further substantiate the robustness of our primary findings, which were obtained through the Bayesian Additive Regression Trees (BART) algorithm, we utilized a neural-sequence learning framework—specifically, the Long Short-Term Memory (LSTM) model—to conduct an additional robustness analysis.

Furthermore, partial dependence plots provide a graphical depiction of the connection between these attributes and the target variable. They unveil the characteristics and intensity of these connections across various values of the predictors.

5.1 Empirical findings for model 1

5.1.1 Model diagnostics for model 1

This section details the model diagnostics, the performance metrics of our predictive model across all banks and a nuanced comparison between big and small banks' lending behaviors (Table 2).

The Bayesian Additive Regression Trees (BART) model was evaluated using datasets from all banks combined, as well as separately for big and small banks. The training set comprised 1,653 observations with 10 predictors, and the testing set consisted of 497 observations with 10 predictors, ensuring substantial data to gauge predictive accuracy and model robustness.

The BART models utilized 250 trees for the combined dataset and 50 trees each for the subsets of big and small banks. Initial (pre-burn-in) and stabilized (post-burn-in) variance estimates indicated minimal fluctuations in predictive accuracy after model stabilization. The parameters alpha (0.95) and beta (1 or 2), determining

Table 2 Model Diagnostics for All, Big and Small Banks

Parameter/Statistic	All		Big		Small	
	Training	Test	Training	Test	Training	Test
num_trees	250	250	50	50	50	50
alpha	0.95	0.95	0.95	0.95	0.95	0.95
beta	1	1	1	1	2	2
k	3	3	2	2	2	2
RMSE	0.020	0.040	0.0353	0.0386	0.0797	0.098
Pre-burn-in sigsq	0.012	0.012	0.004	0.004	0.024	0.00444
Post-burn-in avg sigsq	0.00103	0.0032	0.00044	0.00028	0.0034	0.00188
Pseudo-Rsq	0.9857	0.9493	0.988	0.9824	0.9596	0.9446
Shapiro–Wilk p-val	0	0	0.00014	0	0	0.92757
Zero-mean noise p-val	0.93754	0.8283	0.30426	0.68145	0.96396	0.98853

the prior distribution of tree structures, were carefully selected, alongside the scaling parameter k (2 or 3), to enhance the model's adaptability to underlying data patterns.

Robust in-sample performance was evidenced by low RMSE values across all banks, with training and testing RMSEs of 0.020 and 0.040, respectively, in the combined dataset. Big banks had notably lower RMSE values (training: 0.0353; testing: 0.0386) compared to small banks (training: 0.0797; testing: 0.098), indicating superior prediction precision for large financial institutions. The pseudo-R^2 values (e.g., 0.9857 for training and 0.9493 for testing in the combined dataset) confirmed robust explanatory capabilities, especially notable for larger banks.

Residual analysis using the Shapiro–Wilk test identified deviations from normality (p-values near zero across most subsets), suggesting potential limitations in the model's residual distribution assumptions. Conversely, the zero-mean noise tests consistently validated the accuracy of residual estimation, with p-values close to one.

The BART algorithm handles Markov Chain Monte Carlo (MCMC) simulation in prediction process of the algorithm. Figure 1 illustrates the comprehensive diagnostics overview of MCMC simulation for the dataset of all banks.[3] Figure 1 offers a comprehensive diagnostic overview of these simulations for the combined banks dataset, highlighting variance parameter estimates, acceptance rates, and tree complexity metrics. The convergence of σ^2 estimates, clearly visible in the upper left quadrant with red dots and delineated by the horizontal green lines representing the 95% confidence interval (CI), confirms stable parameter estimates after the burn-in period.

The top right quadrant shows the percentage acceptance rate of the MCMC iterations. After initial fluctuations, the acceptance rate stabilizes, suggesting effective

[3] MCMC simulation results for Big Banks and Small Banks are not presented here to conserve space, but available upon request.

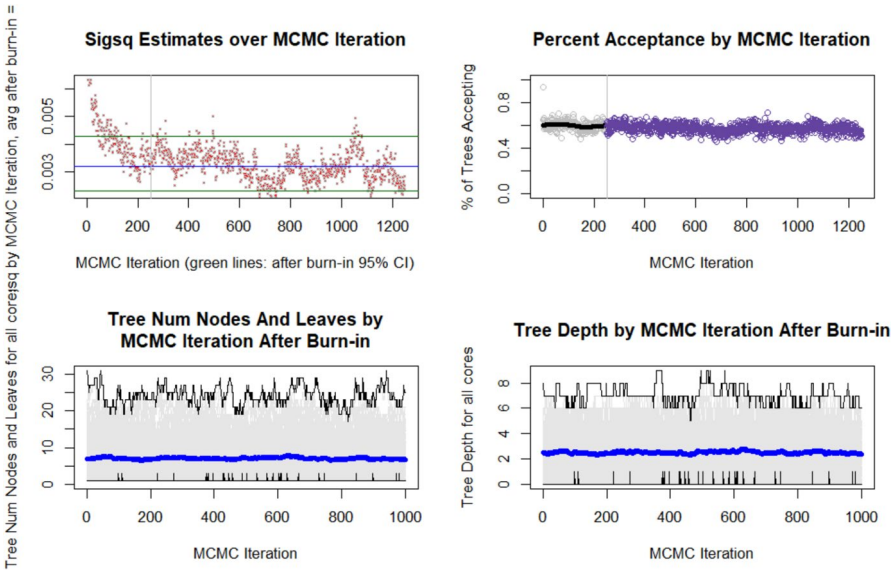

Fig. 1 Markov Chain Monte Carlo (MCMC) Simulation Results

calibration of the algorithm. The lower left quadrant illustrates the number of nodes and leaves in the trees after the burn-in period (Fig. 1).

The blue line indicates a stable number of leaves, while the gray line depicts fluctuations in the number of nodes, reflecting changes in tree complexity over iterations. The lower right quadrant represents the tree depth across MCMC iterations post-burn-in. Here, the consistent blue line indicates stable tree depth, with gray bars showing minor variations.

These diagnostics indicate that the MCMC approach effectively samples from the posterior distribution, maintaining stable variance estimates, consistent acceptance rates and controlled model complexity.

5.1.2 Variable importance results for model 1

In analyzing the determinants of loan dollarization across banking institutions of varying sizes, we observe distinct patterns in variable importance (Fig. 2).

Deposit dollarization and market share emerge as the most influential predictors of foreign currency lending across both large and small banks. However, larger banks show a stronger emphasis on converting deposits into dollars, whereas smaller banks slightly prioritize capturing greater market share. Additionally, the capital ratio and return on assets (roa) exhibit moderate significance across all bank groups, reflecting their role as indicators of financial health influencing dollarization decisions.

Monetary policy rates hold notable importance, particularly among large banks, underscoring their sensitivity to central bank interest rate policies in determining foreign currency lending. In contrast, macroeconomic indicators such as inflation,

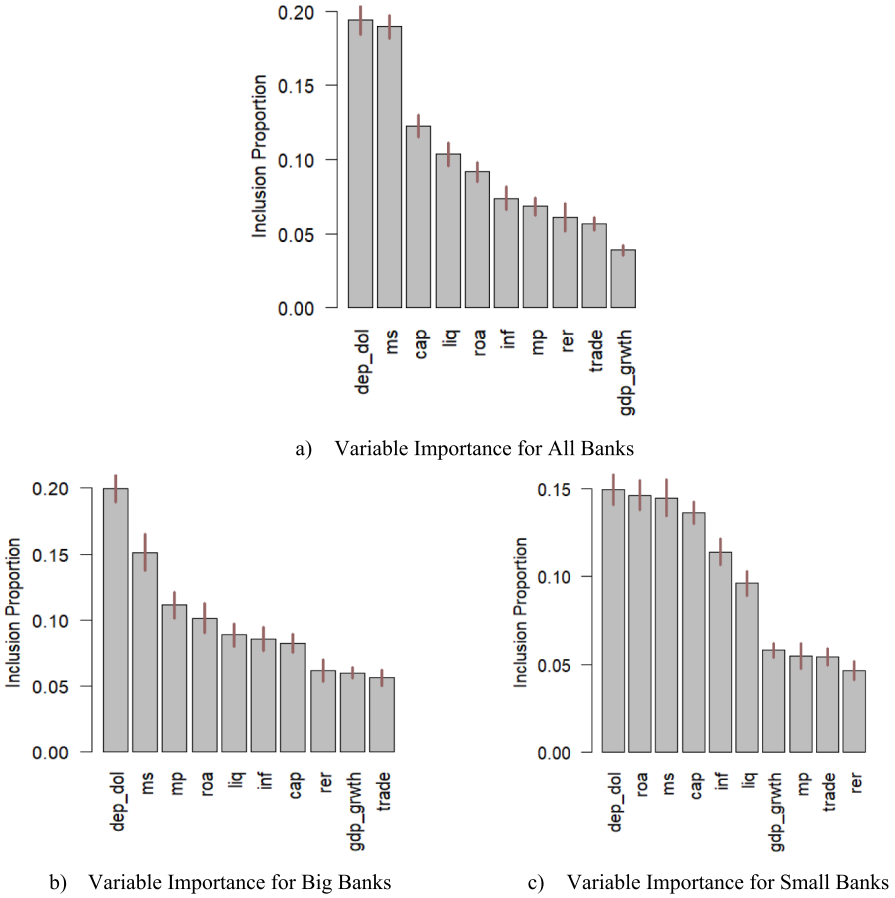

Fig. 2 Variable Importance Results for Model 1

trade openness, real exchange rate, and GDP growth exhibit comparatively lower impact. This suggests that internal banking characteristics and immediate monetary conditions are prioritized over broader economic indicators when banks decide on currency composition for loans.

The findings highlight sophisticated decision-making processes within banks that consider both internal financial positions and external monetary conditions. Banks strategically align asset–liability currencies to mitigate risks, optimize profitability, and comply effectively with regulatory frameworks.

5.1.3 Partial dependence plots for model 1

Partial dependence plots show how variables in the dataset affect the output variable. These plots show the nature and intensity of these correlations across a range of predictor values by methodically documenting the influence of one or

more predictors on the target variable. Therefore, they allow us to see patterns and trends when predictor values change using partial dependence plots, giving a deeper grasp of the model's predictive framework.

Figures 3, 4, and 5 provide partial dependence plots for bank-specific and macroeconomic variables to display how loan dollarization differs for different for lower and upper quantile values of input variables for all banks, big banks and small banks, respectively.

The partial dependence plots (PDPs) indicate that the influence of monetary policy stance on loan dollarization exhibits nonlinear patterns across banks of different sizes. In small banks, an initial rise in monetary policy rates increases foreign currency lending, potentially due to borrowers hedging against higher borrowing costs. In contrast, the initial response of large banks initially to monetary tightening seems to be muted. However, this silence reverses beyond certain interest rate thresholds, highlighting a nonlinear relationship.

The impact of tightening monetary policy on the tendency to borrow in foreign currency can be clarified by examining the notion of interest rate differentials within the context of loan demand and supply. In situations characterized by larger financial institutions, an increase in domestic interest rates triggers demand-side factors, causing economic actors to opt for foreign currency loans. This shift arises when central banks raise domestic interest rates, making loans in foreign currencies relatively cheaper since the difference between the interest rates on domestic and foreign currency loans widens (Basso et al. 2011). This mechanism implies that changes in monetary policy directly impact the appeal of lending in foreign currencies by modifying the relative cost of borrowing across different currencies.

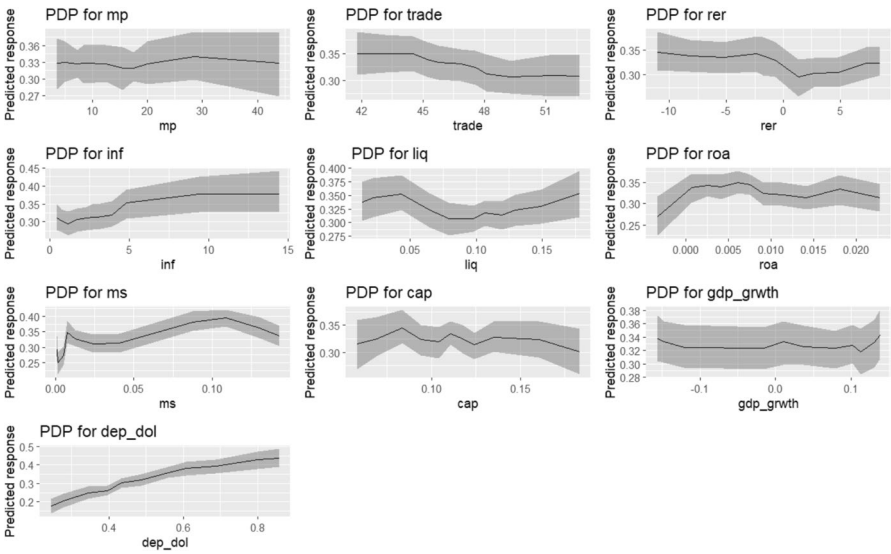

Fig. 3 Partial Dependence Plots for All Banks in Model 1

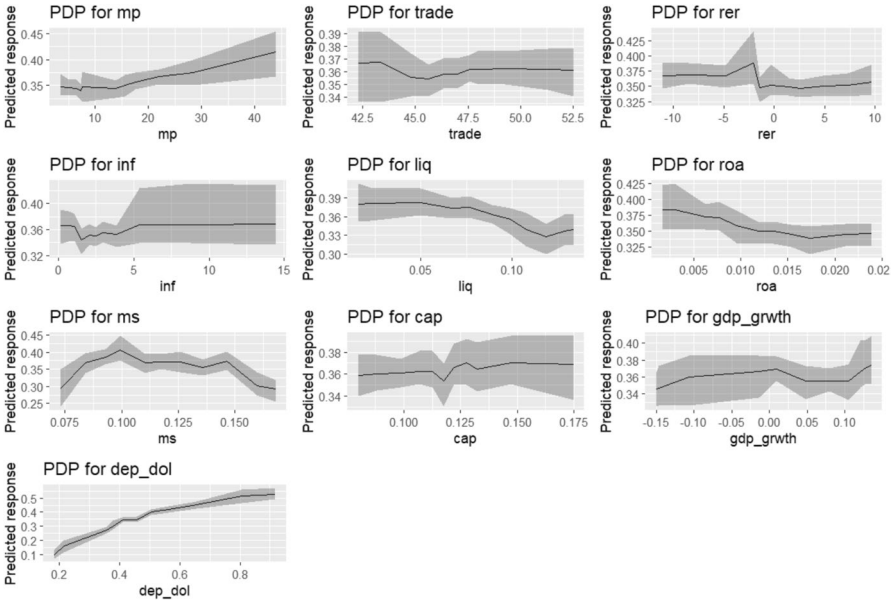

Fig. 4 Partial Dependence Plots for Big Banks in Model 1

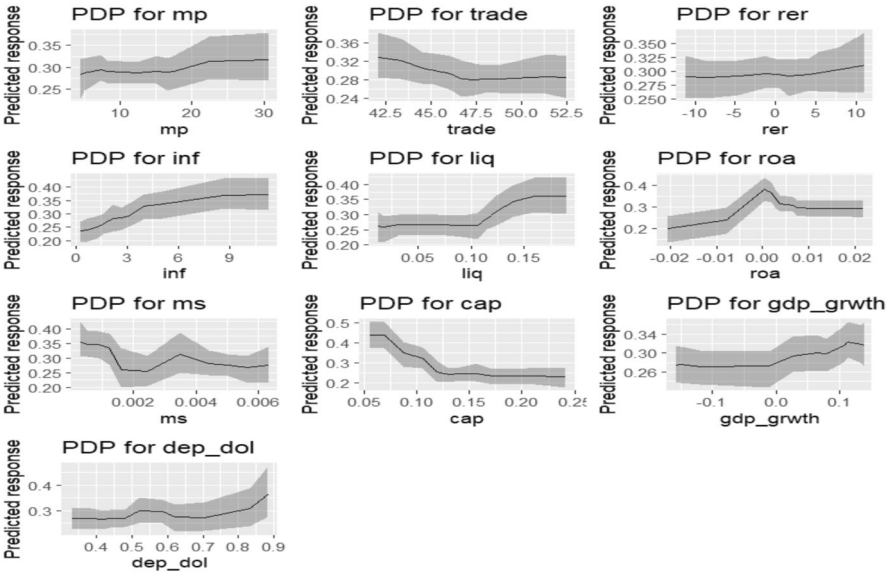

Fig. 5 Partial Dependence Plots for Small Banks in Model 1

However, if supply-side factors dominate the loan market, high domestic interest rates could lead banks to boost their lending in the domestic currency (Castro 2013). Smaller banks tend to be more influenced by supply-side factors, and

when there is a tightening of monetary policy, it leads to a drop in the use of loans denominated in dollars. The implication is that when the loan market conditions are primarily influenced by the availability of credit from lenders, a rise in domestic interest rates acts as an incentive for these institutions to provide more loans in the domestic currency, offsetting the inclination toward borrowing in foreign currency.

The interplay between demand-side and supply-side factors in response to monetary policy changes underscores the complexity of the bank lending composition and its susceptibility to fluctuations in the share of foreign currency loans. The overall effect of the monetary tightening on the proportion of outstanding foreign currency loans in the foreign currency loan market might suggest that the borrower's desire for such loans undermines the efficacy of the monetary stance.

All banks are experiencing a noticeable upward trend, indicating that an increase in the use of dollars for deposits is resulting in a corresponding increase in the use of dollars for loans. Given that banks primarily use deposits to fund their loans (Weller 2000), it is anticipated that an increase in deposit dollarization will lead to a corresponding increase in loan dollarization. Therefore, this study finds evidence supporting "currency-sensitive lending hypothesis" and the results are parallel to findings of Mora (2013) for Mexico and of Özsöz and Algan (2015), Yılmaz (2020) and Çağlayan et al. (2021) for Türkiye. Banks transfer their exchange rate risk to borrowers when they expand their foreign currency lending in proportion to their foreign currency deposits (Neanidis and Savva 2009). Through this action, banks convert their exposure to fluctuations in exchange rates into a potential risk related to the borrower's ability to repay the loan. Currency sensitive lending is more prevalent for large banks. Large banks appear to convert a higher proportion of their foreign currency deposits into foreign currency loans compared to small banks, suggesting that bigger banks have relatively lower exposure to exchange rate risk in their balance sheet positions.

The characteristics of bank balance sheets also support the presence of nonlinear patterns in loan dollarization, both in large and small banks. The PDP for market share has a variable impact on the degree of loan dollarization. At first, the degree to which loans are denominated in dollars rises as a result of a larger market share, maybe because relatively bigger banks are better equipped to handle foreign currencies (Arteta 2005). Özsöz and Algan (2015) utilized a linear model to examine the impact of market concentration on banks' foreign currency lending in Türkiye. They discovered a positive relation between the size of banks and their foreign currency lending. However, our study suggests that this relationship demonstrates a nonlinear characteristic. The impact weakens and even turns around when market share reaches higher levels, possibly due to the diverse range of services offered by larger banks or their capacity to shape market standards.

The plot for the capital ratio exhibits a declining pattern, indicating that banks with higher ratios, which serve as indicators of financial stability and risk, tend to have lower levels of loan dollarization. Well-capitalized banks can absorb losses better, lowering the need for riskier foreign currency lending to maintain profitability. This can alternatively be interpreted as larger banks reducing their vulnerability to currency volatility or adopting a more conservative lending strategy (Arteta 2005).

Banks are the financial firms that are aiming to maximize their profit either through raising funds in domestic or foreign currency. Therefore, the ROA and the loan dollarization nexus might exhibit nonlinear patterns. Banks might boost their foreign currency lending during foreign currency lending is more profitable. Nevertheless, the fluctuation of this pattern is more pronounced for large financial institutions, indicating that they may raise their foreign currency lending when reaching specific levels of return on assets (ROA), potentially in pursuit of greater returns (Figs. 3, 4, and 5).

In the context of trade openness, two conflicting views exist the natural hedging view and the revenue view. Yılmaz (2020) conducted a micropanel econometric study and found evidence supporting the natural hedging view. That is, the study reports that firms in the tradable sector tend to have more foreign currency borrowings compared to the firms in the nontradable sector. For the case of trade openness, in our study for both big and small banks, the trend appears to fluctuate within the specified quantiles, suggesting that the effect of trade openness on loan dollarization is not consistent across different levels of trade openness. The decreasing trend might indicate the access of firms to international markets and reduce their demand for foreign currency loans (Hake et al., 2014). However, the rising trend exhibits that firms increase their level of dollarization to make their transactions (Arteta 2005).

The graph relating the percentage change in real effective exchange rate demonstrates a more significant fluctuation in the partial effect, suggesting that changes in RER may have exact asymmetries during depreciation and appreciation periods. Depreciation periods are linked to increased default risk for borrowers' unhedged loans, and risk-averse institutions would decrease the availability of foreign currency loans. The fluctuation in exchange rates also creates higher exposure to exchange rate risk for borrowers, so they might reduce their demand for foreign currency loans (Steiner 2012).

There is a positive correlation between the inflation rate and loan dollarization, particularly at higher quantiles. This indicates that as inflation rises, its impact on loan dollarization becomes stronger. This phenomenon may be attributed to the inflationary impact on the actual worth of loans and the desire for foreign currency as a safeguard. One possible explanation is that during periods of high inflation, borrowers may be concerned about the central banks implementing measures to raise interest rates. Therefore, they may opt to transition to foreign currency financing (Haiss and Rainer 2012). Finally, the GDP growth stands for controlling the demand in the loan market, and these plots exhibit that economic expansion leads to an increased share of foreign currency loans.

Although there are general patterns, such as the direct correlation between the degree of deposit dollarization and loan dollarization, the strength and instability of these correlations differ depending on the size of the bank. Large financial institutions frequently have more noticeable impacts, possibly because of their wider range of skills and risk management systems. Smaller banks demonstrate greater unpredictability and may be more responsive to shocks, resulting in a more cautious approach to their lending policies in foreign currencies. The disparities are likely a result of the distinct risk profiles, customer bases and strategy priorities of these different bank types.

5.1.4 Interaction plots for model 1

In examining the impact of monetary policy rate on banks' loan dollarization behaviors, significant distinctions emerge between large and small banking institutions, influenced by various factors such as the deposit dollarization ratio, market share, capital ratio and GDP growth.

The interaction plots for Model 1 have been modified to highlight more distinct and clear patterns across bank sizes when assessing the impact of the monetary policy rate on key bank-level determinants on foreign currency lending behavior. These interactions concentrate on the impact of deposit dollarization, market share, capital ratio and GDP growth, offering a perspective on how banks modify their lending strategies in response to changing macrofinancial conditions.

Figure 6 illustrates that loan dollarization often increases in tandem with an increase in the policy rate when deposit dollarization is likewise substantial. This strengthens the "currency-sensitive lending" behavior: Banks that have a larger level of foreign currency funds are more inclined to issue foreign currency loans, especially in tighter monetary situations. Conversely, this effect is less when capital ratios are raised, suggesting that increased capital buffers may reduce risk-taking in foreign currency loans.

Figure 7 illustrates a significant moderating effect: Major banks are inclined to decrease their foreign currency lending during periods of monetary tightening, particularly when their deposit dollarization and capital ratios are elevated. This

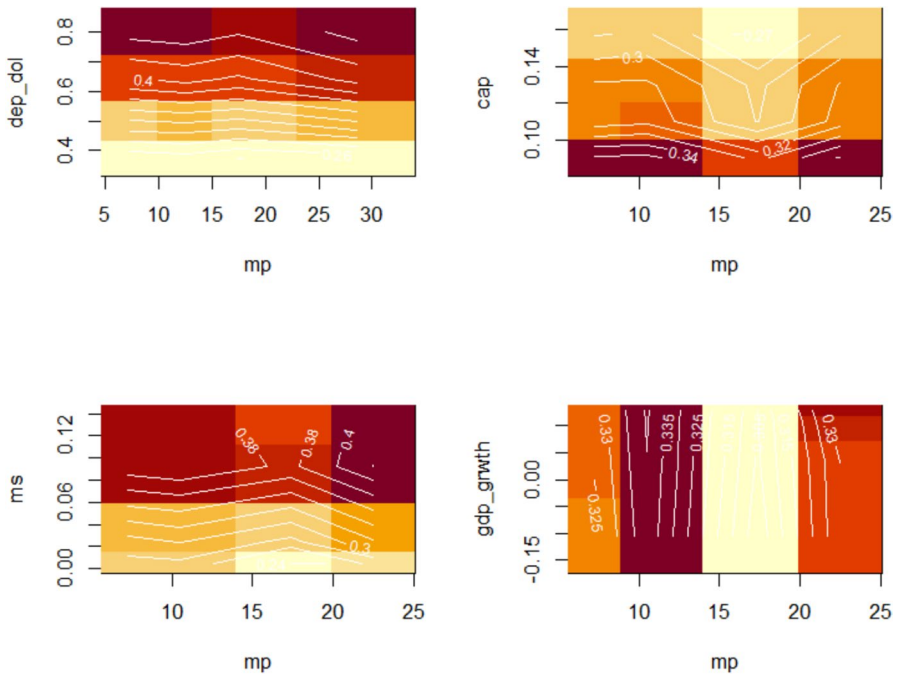

Fig. 6 Interaction Plots for All Banks in Model 1

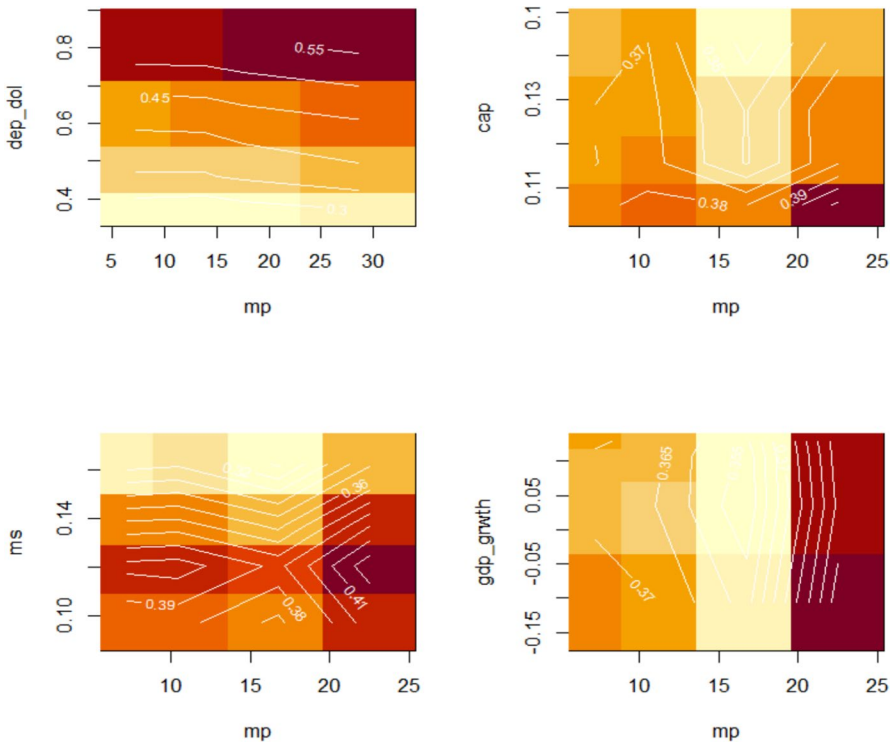

Fig. 7 Interaction Plots for Big Banks in Model 1

implies a conservative attitude to balance sheet risk, which is likely indicative of their wider client base and more regulatory scrutiny. It is intriguing that the interaction with GDP growth demonstrates that large banks somewhat boost their foreign currency lending during expansions. However, this occurs only when capital ratios are low, potentially to leverage short-term benefits (Fig. 7).

A more dynamic and varied pattern is seen for small banks (Fig. 8). Despite the presence of higher policy rates, small banks seem to expand their foreign currency lending when deposit dollarization is high, particularly when their capital ratios are inadequate, or their market share is increasing. This might be indicative of endeavors to maintain profitability in the face of tightening domestic conditions or to compete aggressively in specialized markets. Nevertheless, their lending behavior becomes more cautious during downturns or when capital buffers are large, indicating a heightened vulnerability to systemic shocks.

Smaller banks exhibit adaptive—but riskier—reactions based on their market position and structural restrictions, whereas bigger banks exhibit a more risk-averse and stable profile in tightening conditions. In highly dollarized economies, foreign currency loan remains a crucial route of monetary transmission friction, for which these dynamics are particularly important in the formulation of policy.

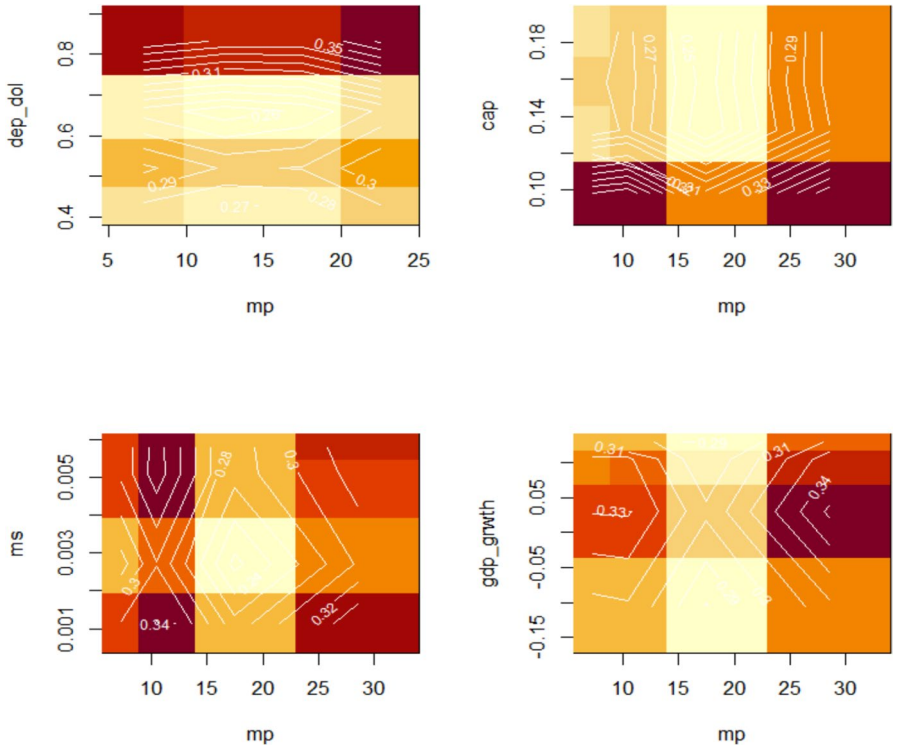

Fig. 8 Interaction Plots for Small Banks in Model 1

5.2 Empirical findings for model 2

In the second model, we look at the expansion of total bank lending in Türkiye's commercial banking sector and take a closer look at how factors like monetary policy and the dollarization of loans affect lending overall. This complete methodology enables us to evaluate not just the immediate impact of monetary policy on the extent to which loans are denominated in dollars, but also its wider consequences for the expansion of bank lending in a changing economic environment.

5.2.1 Model diagnostics for model 2

Table 3 presents the diagnostic statistics for Model 2, which evaluates the determinants of total loan growth using the BART algorithm. The model was trained using 100 trees with hyperparameters alpha and beta values of 0.95 and 2, respectively. This setup encourages a strong ensemble of deep, balanced decision trees, which improves prediction accuracy while avoiding overfitting. The scale parameter k is uniformly set to two, reflecting an ideal balance of model flexibility. The model's efficacy is demonstrated by low RMSE values (0.141 for training and 0.336 for test), which reveal modest disparities between predictions and actuals, particularly with improved generalization in the test dataset. The model's stability is also

Table 3 Model Diagnostics for Model 2

Parameter/Statistic	Training	Test
num_trees	100	100
alpha	0.95	0.95
beta	2	2
k	2	2
RMSE	0.141	0.336
Pre-burn-in sigsq	0.065	0.013
Post-burn-in avg sigsq	0.00332	0.00137
Pseudo-Rsq	0.8939	0.8463

substantiated by the posterior variance estimations. The average posterior variance (sigsq) experienced a substantial decrease from the pre-burn-in values of 0.065 and 0.013 to the post-burn-in levels of 0.00332 (training) and 0.00137 (test). This suggests that the Markov Chain Monte Carlo (MCMC) sampling process was well-calibrated and had converged to stable predictive distributions.

Pseudo-R^2 values are also notably strong, reaching 0.8939 in the training set and 0.8463 in the test set. The BART model covers the bulk of the range in total loan growth, as seen by these numbers, even in out-of-sample projections.

5.2.2 Partial dependence plots for model 2

Partial dependence plots in Fig. 9 display the impact of some selected input variables on the growth of the total loans in commercial banks. The model stands for providing robust evidence favoring the damaging impact of lending in foreign currency on the role of monetary policy on bank loan supply.

The revised partial dependence graphs in Fig. 9 provide a more precise understanding of the impact of certain predictors on the overall loan growth in the Turkish banking industry. The result indicates that the monetary policy rate has an unbalanced and nonlinear impact on the overall increase of loans. In particular, loan growth appears to be generally resistant to minor fluctuations in the policy rate, but it exhibits a significant slowing if the rate crosses specific thresholds. This implies that policy tightening may severely restrict loan supply beyond modest levels, possibly as a result of tighter funding conditions within banks and increasing risk aversion.

The revised figure continues to exhibit a positive link with total loan growth in the lower to midquantiles with respect to loan dollarization. However, this influence becomes less pronounced at higher levels. This saturation suggests that, although foreign currency lending might initially drive credit expansion, its marginal contribution diminishes as dollarization intensifies, presumably due to increased credit risk or regulatory limits.

Conversely, the revised graphic indicates that the impact of deposit dollarization is more nonlinear and moderately concave. Initially, loan growth increases when foreign currency deposit ratios increase, which may indicate a change in financing

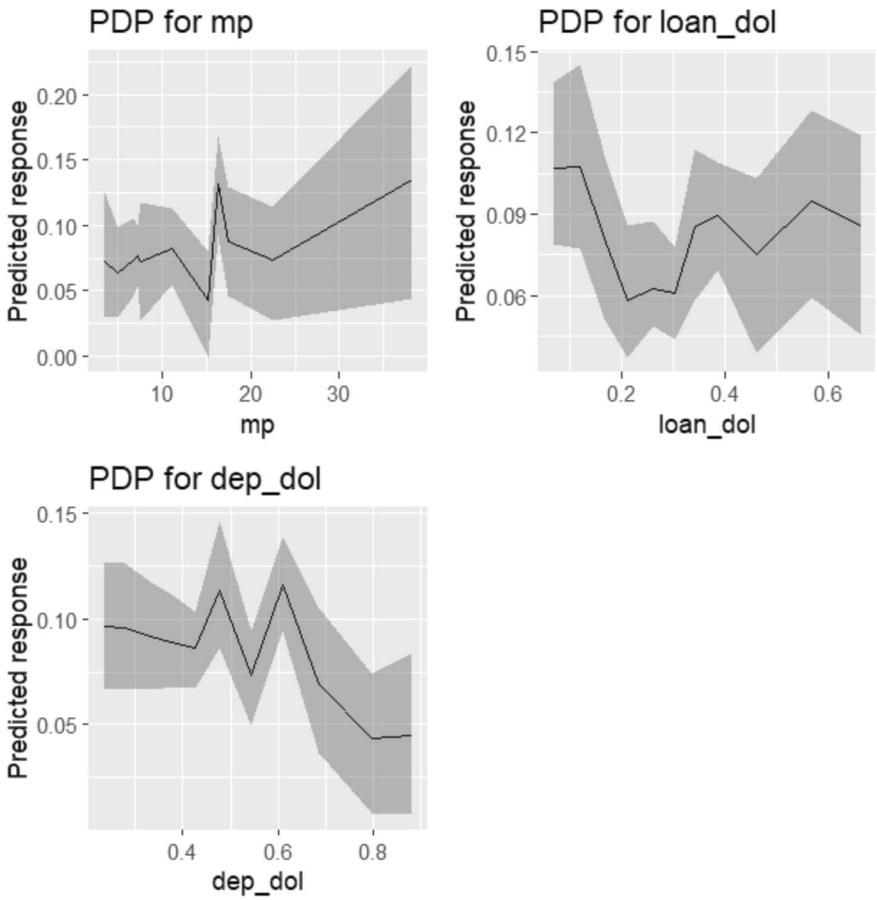

Fig. 9 Partial Dependence Plots for Model 2 (Selected Input Variables)

availability. However, the impact stabilizes or somewhat declines at the higher quantiles. This pattern suggests that banks may use foreign currency deposits to expand lending, but the hedging considerations and balance sheet risks that accompany this practice eventually restrain it.

These findings add validity to the model's conclusion that while lending in foreign currencies may initially enhance loan growth, it may eventually inhibit the main proposition of bank lending channel of monetary policy, especially as dollarization increases.

5.2.3 Interaction plots for model 2

The interaction plots in Fig. 10 depict how monetary policy interacts with loan dollarization and deposit dollarization to affect the growth of total loans in commercial banks.

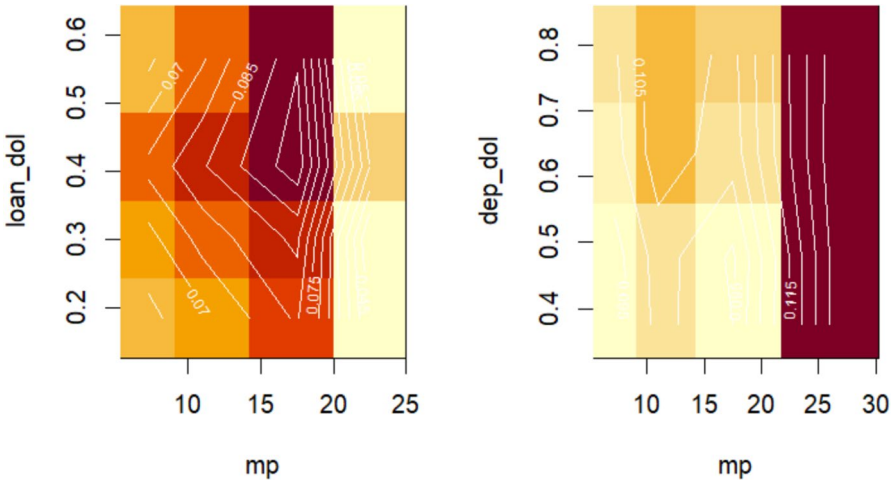

Fig. 10 Interaction Plots for Model 2

The relationship between the policy rate and loan dollarization in the left panel exhibits a nonlinear and asymmetric pattern. Loan growth has a progressive fall as loan dollarization grows at decreasing monetary policy rates. Nevertheless, the deeper coloring indicates that the combined effect of high policy rates and high loan dollarization results in a rapid reduction in loan growth after the policy rate crosses a crucial threshold. This implies that monetary tightening becomes much more restrictive when banks are extensively exposed to foreign currency lending, which supports the idea that loan dollarization exacerbates the contractionary effect of monetary policy on the credit supply.

The right panel illustrates a more flattened and subdued pattern in the interplay between deposit dollarization and monetary policy. Although higher policy rates often result in decreased loan growth, the marginal impact of growing deposit dollarization is less significant than that of loan dollarization. The relatively constant color gradient across various deposit dollarization levels suggests that the influence of monetary policy on credit expansion is more consistent, irrespective of the manner in which banks finance themselves in foreign currency. Nevertheless, the contractionary impacts are exacerbated by even minor increases in deposit dollarization under extreme tightening circumstances.

These graphs emphasize a critical discovery: Loan dollarization has a more pronounced, nonlinear dampening effect on the monetary transmission mechanism than deposit dollarization. This further supports the notion that the responsiveness of banks to domestic monetary policy is dramatically diminished as their loan portfolios grow more dollarized, which presents issues for policy efficacy in highly dollarized economies.

5.3 Robustness check: the long short-term memory (LSTM) model

We used the Long Short-Term Memory (LSTM) model, to strengthen our Bayesian Additive Regression Trees (BART) algorithm findings.[4] This method employs a distinctive modeling approach to evaluate the persistence of the fundamental empirical linkages inside a recurrent deep learning architecture that may capture temporal dependencies.

A subset of deep learning models known as Recurrent Neural Networks (RNNs) is specifically designed for sequential data processing. In this context, observations are chronologically ordered, and present results may be influenced by prior values. Conversely, traditional RNNs are plagued by the disappearing and ballooning gradient issues, which render it challenging to ascertain long-term relationships in time series data (Bengio et al. 1994). Hochreiter and Schmidhuber (1997) proposed the Long Short-Term Memory (LSTM) architecture to address this constraint. This design has subsequently become a fundamental component of financial modeling, natural language processing and time series prediction. LSTM networks are particularly well-suited for incorporating temporal dependencies in macroeconomic and financial variables, where cyclical behavior, policy delays and delayed impacts are crucial. LSTM networks learn such dynamics directly from data using nonlinear function approximation, in contrast to classic econometric models that impose stringent parametric assumptions and predetermined lag structures.

Using a fixed-length input window of four quarters (one year), the LSTM model is trained using quarterly banking panel data in this work, therefore achieving a compromise between economic interpretability and computational efficiency. The network design comprises a dense output layer and a single LSTM layer with 50 hidden units. The model is trained across 50 epochs using the Adam optimizer with a learning rate of 0.001, and the mean squared error (MSE) is tracked to evaluate convergence and prevent overfitting. As a result of this design, the network is capable of predicting the goal variables—loan dollarization and total loan growth—for the subsequent quarter by utilizing the input data from the previous year. Permutation-based approaches are employed to measure the contribution of each predictor to the LSTM's predictive performance via feature significance analysis.

5.3.1 LSTM model results

A single-layer LSTM neural network with a hidden size of 50 and a lookback window of four quarters was built for both Model 1 and Model 2. This decision is indicative of both economic and methodological considerations. A one-year lookback period is an adequate temporal horizon for capturing macroeconomic cycles, policy delays and banking activity, while upholding model parsimony. In order to guarantee convergence and prevent overfitting, the model was trained for 50 epochs using the Adam optimizer

[4] We gratefully acknowledge the anonymous referee for suggesting the incorporation of a robustness check using the LSTM model, which has significantly enhanced the empirical depth and reliability of our analysis.

(learning rate = 0.001). The mean squared error (MSE) was tracked during the training process.

Each input to the model is a 4 × N matrix, where N is the number of predictors used: 9 for Model 1 (e.g., monetary policy rate, deposit dollarization, market share) and 10 for Model 2. (including loan dollarization). In the subsequent quarter, the dependent variable is the goal variable—loan dollarization for Model 1 and total loan growth for Model 2.

The training loss of the LSTM for Model 1 decreased consistently from 0.2271 to 0.0157 during 50 epochs, indicating that it attained high performance. The model's ability to capture the same nonlinear patterns identified with BART was confirmed by an RMSE of 0.1206 on the test set.

A permutation-based feature significance analysis demonstrated that the model's predictions were most significantly influenced by deposit dollarization (dep dol), with trade openness and capital ratio following in that order. Conversely, macroeconomic restrictions, including inflation and the policy interest rate, had minimal or slightly negative impacts on forecast accuracy. The BART variable relevance results for Model 1 are largely consistent with these findings, which emphasize that foreign currency deposit ratios and bank market structure are the primary determinants of loan dollarization in Turkey.

The predicted consistency across methods was validated by the LSTM model trained with comparable parameters, which yielded a test RMSE of 0.1503, which closely aligned with the BART-based RMSE (0.2365) for Model 2.

The most significant determinants of credit growth, according to the feature significance results for Model 2, are the capital ratio and return on assets (ROA). The importance of dollarization measures (loan_dol and dep_dol) was also maintained, which is consistent with the findings of BART. This conclusion indicates that the efficacy of monetary policy transmission through the bank lending channel is weakened by greater dollarization. In the interim, the supremacy of bank-specific traits was further reinforced by the limited influence of traditional macrocontrols, like inflation, trade openness and liquidity.

Our findings are robust to alternative methods—both BART and LSTM yield consistent results. Deposit dollarization, bank capital and profitability emerge in both models as the key drivers of foreign currency lending and credit growth. Once we control for bank fundamentals, inflation and policy rates lose significance, implying that structural bank characteristics dominate monetary policy effects under sustained dollarization (Figs. 11, 12).

Particularly in policymaking scenarios where interpretability and predictive performance must be concurrently assessed, the convergence of results from a nonparametric tree-based model and a deep sequential neural network increases confidence in the validity and generalizability of the findings.

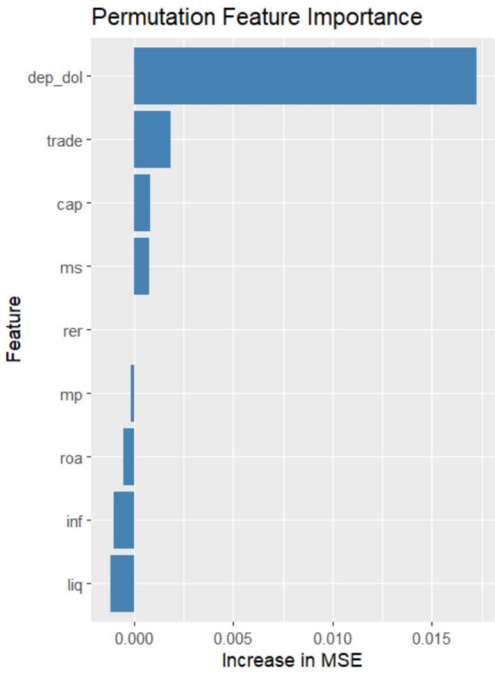

Fig. 11 LSTM Variable Importance Results for Model 1

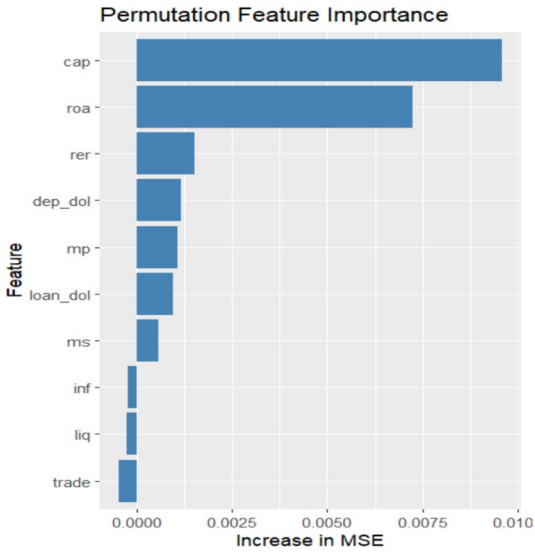

Fig. 12 LSTM Variable Importance Results for Model 2

6 Discussion of findings

One of the key insights from our empirical analysis concerns the mechanisms through which monetary policy affects credit dollarization behavior at the bank

level. As shown in the partial dependence plots of Model 1 (Figs. 3, 4, 5), the responsiveness of dollarization to interest rate changes varies significantly by bank size. Larger banks, with greater balance sheet flexibility and risk management capacity, tend to shift lending toward foreign currency in response to rising policy rates, reflecting strategic portfolio adjustments rather than credit contraction. In contrast, smaller banks constrained by liquidity and capital respond more cautiously.

This behavioral asymmetry forms a structural channel that undermines monetary transmission. Rather than uniformly restricting credit, rate hikes trigger shifts in the currency composition of lending, weakening the domestic credit channel and complicating stabilization efforts. These findings align with Calvo and Reinhart's (2002) view that currency substitution distorts monetary signals in financially dollarized systems. Our results extend this logic into a practical context, illustrating how institutional incentives to reallocate portfolios under dollarization reinforce policy ineffectiveness.

Partial dependence plots further reveal that the response of credit dollarization to policy rates is nonlinear—policy shifts only affect lending behavior once certain thresholds are crossed, such as heightened liquidity stress or inflation expectations. This supports the use of flexible, nonparametric models like BART, as linear approaches may miss such threshold effects.

These asymmetries are deeply shaped by Turkey's broader macroeconomic environment. Since 2016, persistent inflation, exchange rate volatility and waning monetary-policy credibility have intensified dollarization pressures. The 2018 Brunson crisis and the ultra-loose stance of the Türkiye Economy Model adopted in 2021—maintaining low policy rates despite rising inflation—further eroded the signaling power of interest rates (Ulug et al. 2023).

In this context, both banks and borrowers have rationally shifted toward foreign currency to hedge against domestic instability. Our model confirms that credit dollarization behavior is structurally reinforced under such conditions. As macroeconomic volatility persists, banks increasingly prioritize foreign currency lending, further diluting the contractionary effects of monetary policy (Ongena et al. 2020).

To restore monetary traction, policymakers must complement interest rate tools with targeted macroprudential measures, such as FX-based capital buffers and differentiated reserve requirements (Levy Yeyati 2006). However, technical instruments alone are insufficient. As Ulug et al. (2023) stress, successful monetary transmission in a dollarized economy depends critically on institutional coordination, credibility and consistent policy communication.

In Türkiye, the weakening of central bank independence and inconsistent signaling—particularly after the 2021 rate cuts—has eroded monetary credibility, prompting banks to prioritize internal risk strategies over policy alignment. This reflects a deeper institutional misalignment: Without stable, rules-based guidance, even sophisticated macroprudential interventions may fail to anchor expectations (Başçı, Kara, & Öğünç, 2007).

Our findings reinforce the view that ensuring effective monetary transmission under financial dollarization requires not only sound technical tools but also a coherent, credible and autonomous policy orientation. This section offers an integrated interpretation of the results across four dimensions: mechanisms, macroeconomic

background, policy implications and institutional context, using the flexible, nonlinear modeling power of BART to uncover structural constraints that conventional models might overlook.

7 Conclusion

This study examines the relationship between monetary policy adjustments by the CBRT and banks' foreign currency lending behavior in a highly dollarized financial system. Using the BART model operating within ML framework, the research covers the 2002Q4-2024Q3 period and provides detailed empirical analysis.

The findings reveal that the impact of monetary policy on foreign currency lending differs between large and small banks. Larger banks tend to reduce their foreign currency lending in response to tighter monetary policy, influenced by factors such as deposit dollarization and capital ratio. In contrast, smaller banks often expand their foreign currency lending under similar conditions, potentially driven by profit margins or consumer demand. However, this behavior is influenced by market share and capital ratio, indicating a more complex response to monetary policy changes.

Crucially, the study identifies deposit dollarization, market share and return on assets (ROA) as the key determinants of loan dollarization behavior. Deposit dollarization, in particular, emerges as a dominant driver, reinforcing the "currency-sensitive lending hypothesis" whereby banks match their liabilities and assets to mitigate currency mismatch risks. Higher market share and stronger profitability metrics also shape banks' responsiveness to monetary shocks, reflecting institutional differences in balance sheet management strategies.

Capital adequacy emerges as a critical moderating factor in loan dollarization, with well-capitalized banks displaying a lower propensity to engage in foreign currency lending. In addition, trade openness exerts heterogeneous effects across different quantiles, suggesting that external sector dynamics interact with banking behavior in complex ways. Importantly, the analysis reveals pronounced nonlinearities and threshold effects: The influence of monetary policy on lending behavior becomes significant only once certain macrofinancial stress levels are surpassed. These findings highlight the inherent limitations of conventional linear models and underscore the necessity of employing flexible ML techniques capable of capturing such hidden patterns. These empirical findings are also supported by the robustness check through a novel ML algorithm.

Overall, this research provides a nuanced understanding of how central bank policy changes affect banks' foreign currency lending in a dollarized economy. It highlights the need for tailored regulatory and policy frameworks that consider the responses of different bank sizes and characteristics to monetary policy adjustments. The study's use of the BART-ML model and its focus on loan dollarization and total loan growth offer valuable insights for policymakers and financial regulators aiming to enhance the effectiveness of monetary interventions and ensure financial stability in dollarized economies.

Overall, the Turkish experience serves as a cautionary tale for other emerging economies grappling with the challenges of persistent dollarization. In environments

where structural vulnerabilities are deep and institutional credibility is eroding, monetary policy alone may prove insufficient to stabilize credit markets. Addressing these challenges demands not only technical precision but also institutional resilience and coordinated policy actions across the financial and fiscal domains.

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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